

1 Summary

Model (Run-name): model3 (test)

Iteration: 8

TAC: 63291.16746090113

Took: 87.55484509468079s

Total Time: 291.0738773345947s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html

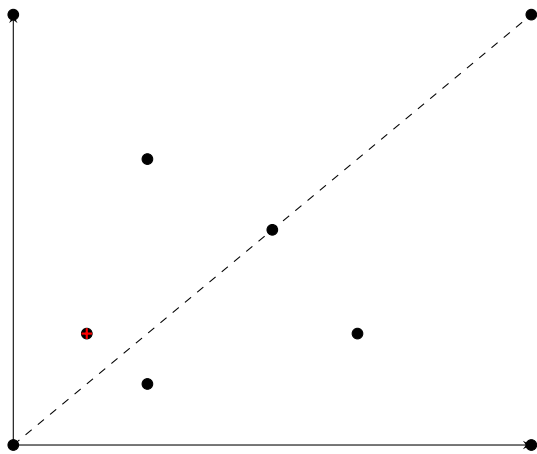
2 Results

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	1.0	$dt_{2,5,2}$	28.17547169811357	$q_{1,1,2}$	586.96
$z_{2,1,1}$	1.0	$dt_{2,5,3}$	44.5	$q_{2,1,1}$	173.039999999998723
$z_{2,2,2}$	1.0	$dt_{3,5,2}$	22.3	$q_{2,2,2}$	647.2099999999999
$z_{2,5,2}$	1.0	$dt_{3,5,3}$	53.205405405405386	$q_{2,5,2}$	357.41000000000395
$z_{3,5,2}$	1.0	$dt_{4,3,2}$	50.0	$q_{3,5,2}$	1186.8799999999965
$z_{4,3,2}$	1.0	$dt_{4,3,3}$	111.1	$q_{4,3,2}$	1539.7200000000003
$z_{5,4,1}$	1.0	$dt_{5,4,1}$	22.2	$q_{5,4,1}$	1634.85
$z_{cu,3}$	1.0	$dt_{5,4,2}$	24.335593220338975	$q_{cu,3}$	1197.4
$z_{cu,5}$	1.0	$dt_{cu,3}$	106.50540540540538	$q_{cu,5}$	724.5600000000002
$A_{1,1,2}^\beta$	6.705169702016156	$dt_{cu,5}$	66.535593220339	$q_{1,1}^{(H)}$	160.0
$A_{2,1,1}^\beta$	0.6681872659240857	$f_{1,1,2}^H$	8.8	$q_{1,2}^{(H)}$	160.0
$A_{2,2,2}^\beta$	8.84227992538678	$f_{2,1,1}^H$	1.5575157515750868	$q_{1,3}^{(H)}$	93.3
$A_{2,5,2}^\beta$	1.5504287284402025	$f_{2,2,2}^H$	7.382988298829847	$q_{2,1}^{(H)}$	248.9
$A_{3,5,2}^\beta$	6.746672991462779	$f_{2,5,2}^H$	3.217011701170153	$q_{2,2}^{(H)}$	232.57547169811357
$A_{4,3,2}^\beta$	6.675811553971783	$f_{3,5,2}^H$	7.367349472377384	$q_{2,3}^{(H)}$	137.8
$A_{5,4,1}^\beta$	13.119332702612118	$f_{4,3,2}^H$	12.6	$q_{3,1}^{(H)}$	226.7
$A_{cu,3}^\beta$	5.6687875135985	$f_{5,4,1}^H$	12.264441110277563	$q_{3,2}^{(H)}$	226.7
$A_{cu,5}^\beta$	5.308038252896679	$f_{1,1,2}^C$	5.86960000000002	$q_{3,3}^{(H)}$	146.50540540540538
$A_{1,1,2}$	24.928068483288385	$f_{2,1,1}^C$	7.6	$q_{4,1}^{(H)}$	271.1
$A_{2,1,1}$	1.0776943916793027	$f_{2,2,2}^C$	6.1	$q_{4,2}^{(H)}$	271.1
$A_{2,2,2}$	41.49476194796521	$f_{2,5,2}^C$	3.217011701170153	$q_{4,3}^{(H)}$	148.9
$A_{2,5,2}$	2.702328746408619	$f_{3,5,2}^C$	10.682988298829848	$q_{5,1}^{(H)}$	198.9
$A_{3,5,2}$	27.976182369080462	$f_{4,3,2}^C$	8.4	$q_{5,2}^{(H)}$	106.535593220339
$A_{4,3,2}$	23.668654734065058	$f_{5,4,1}^C$	17.3	$q_{5,3}^{(H)}$	106.535593220339
$A_{5,4,1}$	82.70899711080247	$RecLMTD_{1,1,2}$	0.03609932228907442	$q_{1,1}^{(C)}$	160.0
$A_{cu,3}$	18.852085704732907	$RecLMTD_{2,1,1}$	0.014696478583585155	$q_{1,2}^{(C)}$	137.2315789473701
$A_{cu,5}$	16.235135805353295	$RecLMTD_{2,2,2}$	0.05449629587888078	$q_{1,3}^{(C)}$	60.0
$dt_{1,1,2}$	22.768421052630092	$RecLMTD_{2,5,2}$	0.02659374531041192	$q_{2,1}^{(C)}$	221.7
$dt_{1,1,3}$	33.3	$RecLMTD_{3,5,2}$	0.022175692011985354	$q_{2,2}^{(C)}$	221.7
$dt_{2,1,1}$	88.9	$RecLMTD_{4,3,2}$	0.013066243553344306	$q_{2,3}^{(C)}$	115.6
$dt_{2,1,2}$	50.610873882818595	$RecLMTD_{5,4,1}$	0.04300250637317342	$q_{3,1}^{(C)}$	221.1
$dt_{2,2,2}$	10.875471698113586	$RecLMTD_{cu,3}$	0.01463328038407244	$q_{3,2}^{(C)}$	221.1
$dt_{2,2,3}$	22.2	$RecLMTD_{cu,5}$	0.01904585601544449	$q_{3,3}^{(C)}$	37.8

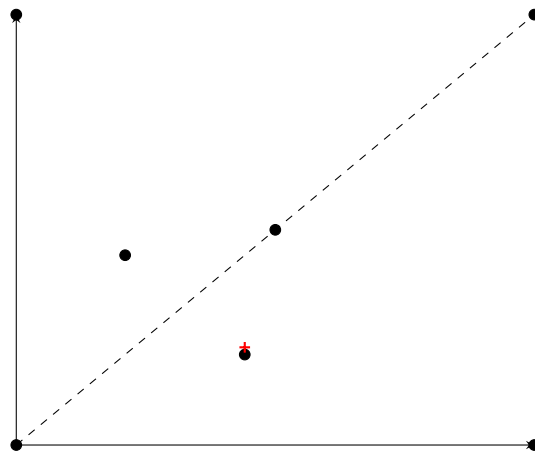
Variable	Value	Variable	Value
$t_{4,1}^{(C)}$	176.7	$b_{5,4,1}^{H,out}$	804.5473368342082
$t_{4,2}^{(C)}$	82.2	$b_{1,1,2}^{C,in}$	352.1760000000001
$t_{4,3}^{(C)}$	82.2	$b_{2,1,1}^{C,in}$	1042.9600000000128
$t_{5,1}^{(C)}$	204.4	$b_{2,2,2}^{C,in}$	705.16
$t_{5,2}^{(C)}$	204.4	$b_{2,5,2}^{C,in}$	300.1471917191753
$t_{5,3}^{(C)}$	93.3	$b_{3,5,2}^{C,in}$	996.7228082808248
$t_{1,1,2}^H$	93.3	$b_{4,3,2}^{C,in}$	317.52
$t_{2,1,1}^H$	139.290053254468	$b_{5,4,1}^{C,in}$	1422.0600000000002
$t_{2,2,2}^H$	137.8	$b_{1,1,2}^{C,out}$	939.1360000000002
$t_{2,5,2}^H$	137.8	$b_{2,1,1}^{C,out}$	1216.0
$t_{3,5,2}^H$	94.10325108061414	$b_{2,2,2}^{C,out}$	1352.37
$t_{4,3,2}^H$	148.9	$b_{2,5,2}^{C,out}$	657.5571917191792
$t_{5,4,1}^H$	65.6	$b_{3,5,2}^{C,out}$	2183.602808280821
$t_{1,1,2}^C$	160.0	$b_{4,3,2}^{C,out}$	1857.2400000000002
$t_{2,1,1}^C$	160.0	$b_{5,4,1}^{C,out}$	3056.91
$t_{2,2,2}^C$	221.7		
$t_{2,5,2}^C$	148.34264863679172		
$t_{3,5,2}^C$	204.4		
$t_{4,3,2}^C$	221.1		
$t_{5,4,1}^C$	176.7		
$b_{1,1,2}^{H,in}$	1408.0		
$b_{2,1,1}^{H,in}$	387.6656705670391		
$b_{2,2,2}^{H,in}$	1664.5857875787528		
$b_{2,5,2}^{H,in}$	800.714212421251		
$b_{3,5,2}^{H,in}$	1670.1781253879528		
$b_{4,3,2}^{H,in}$	3415.8599999999997		
$b_{5,4,1}^{H,in}$	2439.3973368342076		
$b_{1,1,2}^{H,out}$	821.0400000000002		
$b_{2,1,1}^{H,out}$	214.62567056705188		
$b_{2,2,2}^{H,out}$	1017.375787578753		
$b_{2,5,2}^{H,out}$	443.30421242124703		
$b_{3,5,2}^{H,out}$	483.29812538795636		
$b_{4,3,2}^{H,out}$	1876.1399999999996		

3 *RecLMTD* Tangent Points

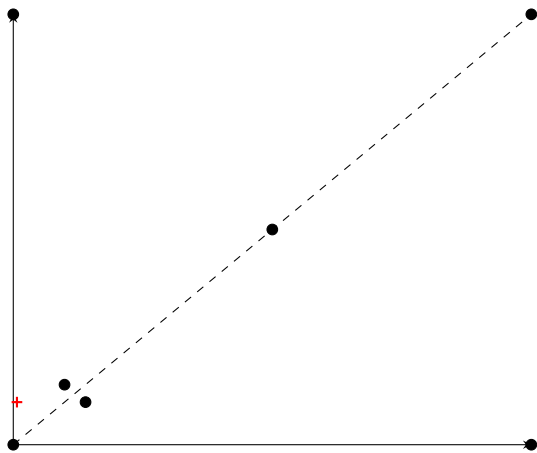
- (1, 1, 2)
 - Value: 0.036099
 - Error: 0.0000000000000001
 - Relative error: 0.0000000000000021
- (2, 1, 1)
 - Value: 0.014696
 - Error: 0.000016463381188
 - Relative error: 0.001118972753864
- (2, 2, 2)
 - Value: 0.054496
 - Error: 0.008515806637220
 - Relative error: 0.135145571996177
- (2, 5, 2)
 - Value: 0.026594
 - Error: 0.001403228274930
 - Relative error: 0.050120712892505
- (3, 5, 2)
 - Value: 0.022176
 - Error: 0.005960917318262
 - Relative error: 0.211856277645156
- (4, 3, 2)
 - Value: 0.013066
 - Error: 0.000000985435484
 - Relative error: 0.000075412735578
- (5, 4, 1)
 - Value: 0.043003
 - Error: 0.000005558310536
 - Relative error: 0.000129238796873
- $cu, 3$
 - Value: 0.014633
 - Error: 0.000000233609228
 - Relative error: 0.000015963987054
- $cu, 5$
 - Value: 0.019046
 - Error: 0.000000131098315
 - Relative error: 0.000006883251277



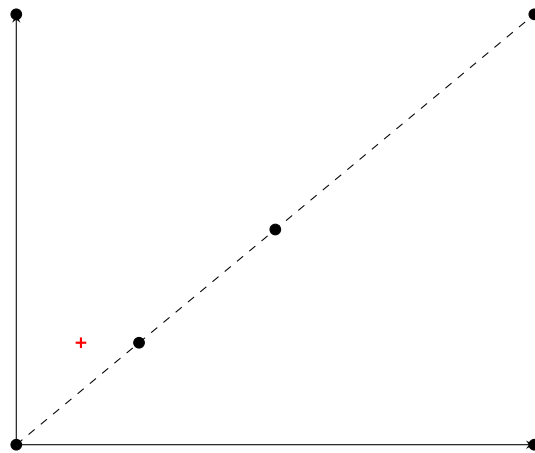
(a) *RecLMTD* - (1, 1, 2)



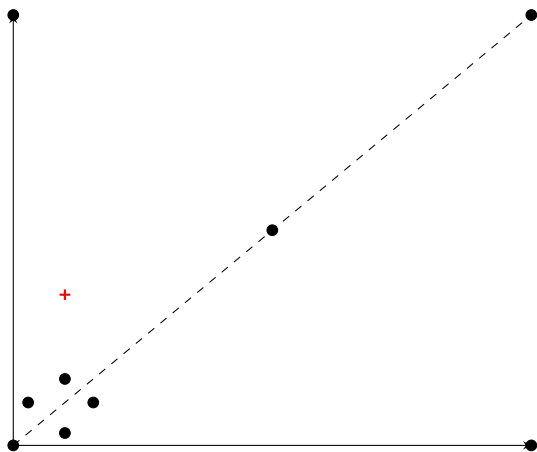
(b) *RecLMTD* - (2, 1, 1)



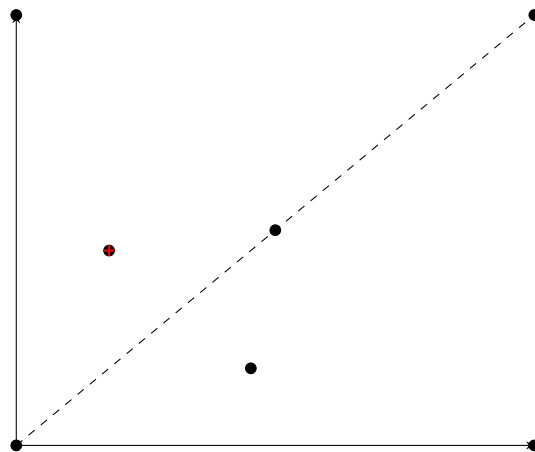
(c) *RecLMTD* - (2, 2, 2)



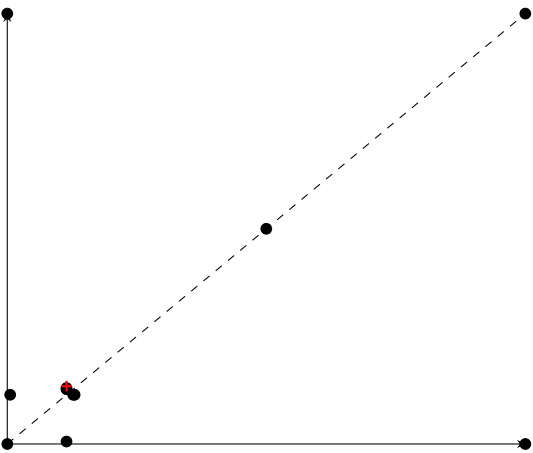
(d) *RecLMTD* - (2, 5, 2)



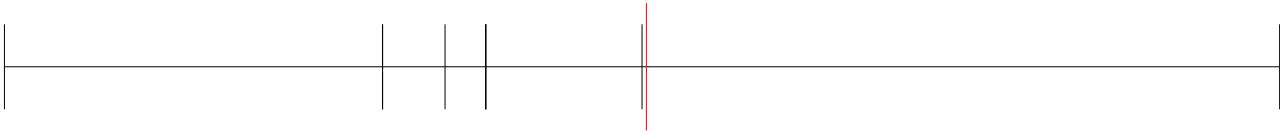
(e) *RecLMTD* - (3, 5, 2)



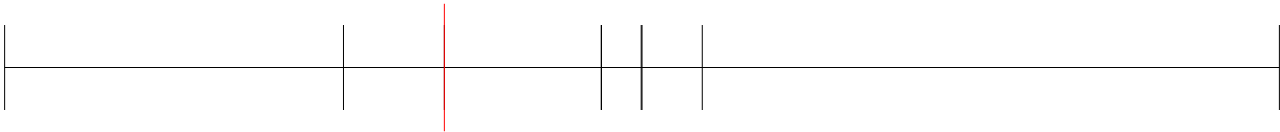
(f) *RecLMTD* - (4, 3, 2)



(g) *RecLMTD* - (5, 4, 1)



(a) *RecLMTD* - $cu, 3$



(b) *RecLMTD* - $cu, 5$

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 2)
 - Value: 1408.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 1)
 - Value: 387.6656705670391
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 2)
 - Value: 1664.5857875787528
 - Absolute error: 52.51619856325192
 - Relative error: -0.03058420465824842
- (2, 5, 2)
 - Value: 800.714212421251
 - Absolute error: 52.51619856325192
 - Relative error: 0.0701902405386751
- (3, 5, 2)
 - Value: 1670.1781253879528
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 3415.8599999999997
 - Absolute error: 4.547473508864641e-13
 - Relative error: -1.3312821687260722e-16
- (5, 4, 1)
 - Value: 2439.3973368342076
 - Absolute error: 4.547473508864641e-13
 - Relative error: 1.8641790905479326e-16

4.2 $b^{H,\text{out}}$

- (1, 1, 2)
 - Value: 821.0400000000002
 - Absolute error: 1.1368683772161603e-13
 - Relative error: 1.3846686851020172e-16
- (2, 1, 1)

- Value: 214.62567056705188
 - Absolute error: 2.3207814145147267
 - Relative error: -0.010697484993725168
- (2, 2, 2)
 - Value: 1017.375787578753
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 5, 2)
 - Value: 443.30421242124703
 - Absolute error: 5.684341886080802e-14
 - Relative error: -1.282266607626839e-16
- (3, 5, 2)
 - Value: 483.29812538795636
 - Absolute error: 209.99341180980275
 - Relative error: -0.3028933724744181
- (4, 3, 2)
 - Value: 1876.1399999999996
 - Absolute error: 4.547473508864641e-13
 - Relative error: -2.4238455066597593e-16
- (5, 4, 1)
 - Value: 804.5473368342082
 - Absolute error: 1.1368683772161603e-13
 - Relative error: 1.41305343410817e-16

4.3 $b^{C,in}$

- (1, 1, 2)
 - Value: 352.17600000000001
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 1)
 - Value: 1042.96000000000128
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 2)
 - Value: 705.16
 - Absolute error: 0.0
 - Relative error: 0.0

- (2, 5, 2)
 - Value: 300.1471917191753
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 2)
 - Value: 996.7228082808248
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 317.52
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 1)
 - Value: 1422.0600000000002
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 2)
 - Value: 939.1360000000002
 - Absolute error: 1.1368683772161603e-13
 - Relative error: -1.2105471169416996e-16
- (2, 1, 1)
 - Value: 1216.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 2)
 - Value: 1352.37
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 5, 2)
 - Value: 657.5571917191792
 - Absolute error: 180.3371552720476
 - Relative error: 0.37789099681280064
- (3, 5, 2)
 - Value: 2183.602808280821
 - Absolute error: 0.0

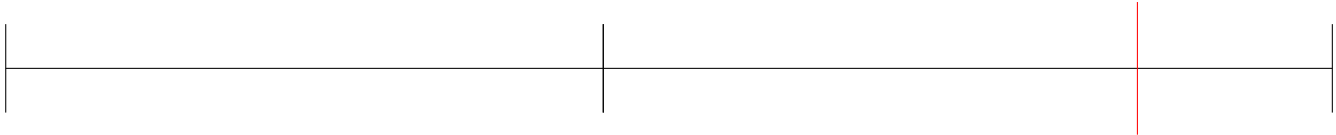
- Relative error: 0.0
- (4, 3, 2)
 - Value: 1857.2400000000002
 - Absolute error: 2.2737367544323206e-13
 - Relative error: 1.2242557528549464e-16
- (5, 4, 1)
 - Value: 3056.91
 - Absolute error: 0.0
 - Relative error: 0.0



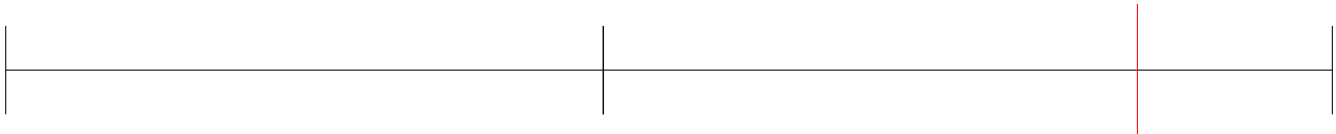
(a) $t^{(H)} - (1, 2)$



(a) $t^{(H)} - (2, 1)$



(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (3, 2)$



(a) $t^{(H)} - (4, 2)$



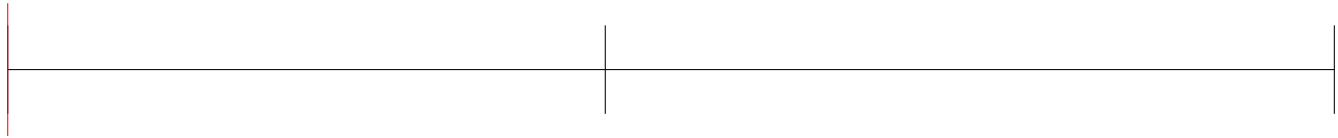
(a) $t^{(H)} - (5, 1)$



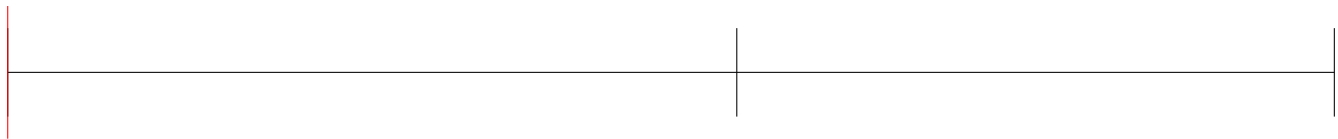
(a) $t^{(H)} - (1, 1, 2)$



(a) $t^{(H)} - (2, 1, 1)$



(a) $t^{(H)} - (2, 2, 2)$



(a) $t^{(H)} - (2, 5, 2)$



(a) $t^{(H)} - (3, 5, 2)$



(a) $t^{(H)} - (4, 3, 2)$



(a) $t^{(H)} - (5, 4, 1)$



(a) $t^{(C)} - (1, 3)$



(a) $t^{(C)} - (1, 2)$



$$(a) \ t^{(C)} - (2, 3)$$



$$(a) \ t^{(C)} - (5, 3)$$



$$(a) \ t^{(C)} - (5, 3)$$



$$(a) \ t^{(C)} - (3, 3)$$



$$(a) \ t^{(C)} - (4, 2)$$



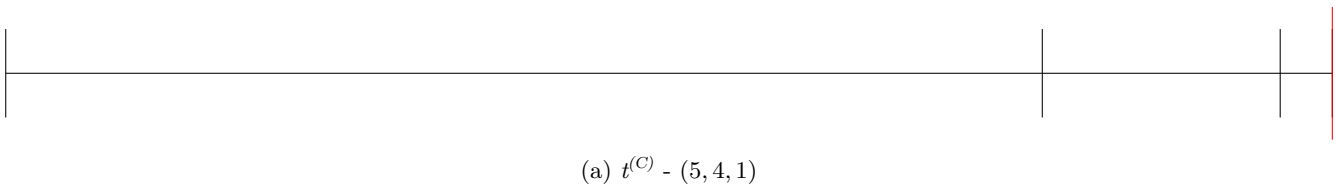
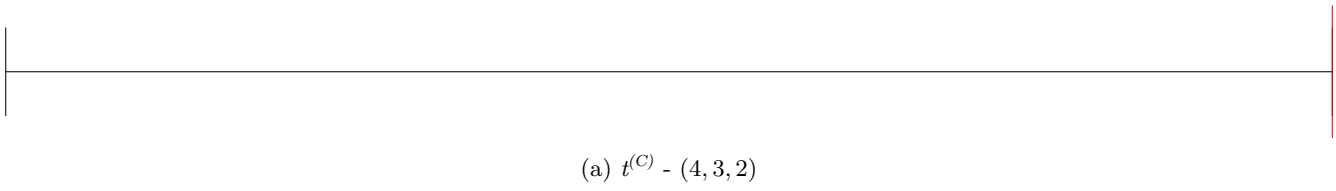
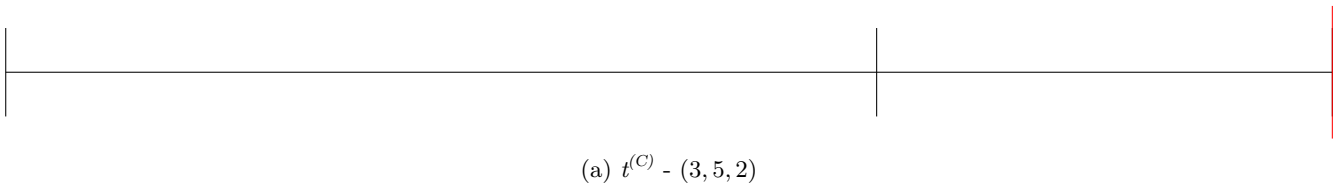
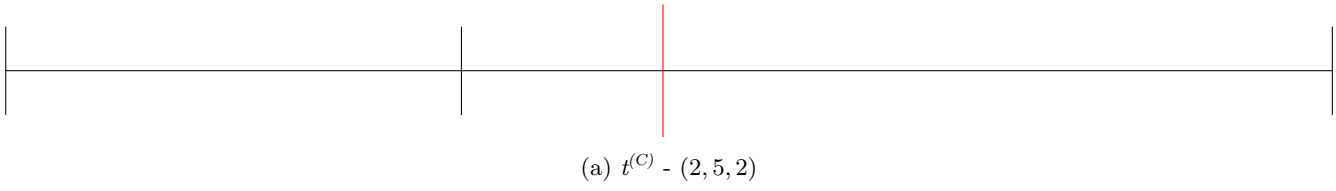
$$(a) \ t^{(C)} - (1, 1, 2)$$



$$(a) \ t^{(C)} - (2, 1, 1)$$



$$(a) \ t^{(C)} - (2, 2, 2)$$



5 Area Breakpoints

5.1 A

- $(1, 1, 2)$
 - Value: 24.928068
 - Error: 0.0000000000000004
 - Relative error: -0.0000000000000000
- $(2, 1, 1)$
 - Value: 1.077694
 - Error: 1.914162848442330
 - Relative error: 0.639790837200678
- $(2, 2, 2)$
 - Value: 41.494762
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $(2, 5, 2)$
 - Value: 2.702329
 - Error: 8.479871855231888
 - Relative error: 0.758336588416043
- $(3, 5, 2)$
 - Value: 27.976182
 - Error: 2.988388613490240
 - Relative error: 0.096509931145900
- $(4, 3, 2)$
 - Value: 23.668655
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $(5, 4, 1)$
 - Value: 82.708997
 - Error: 0.0000000000000540
 - Relative error: 0.0000000000000007

5.2 A_{CU}

- $cu, 3$
 - Value: 18.852086
 - Error: 1.761902450429851
 - Relative error: 0.085471207083651
- $cu, 5$

- Value: 16.235136
- Error: 0.000000000000192
- Relative error: 0.000000000000012



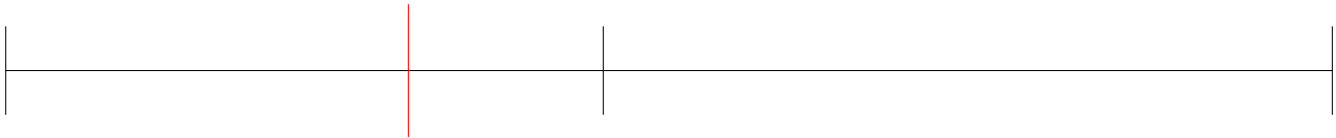
(a) $A^\beta - (1, 1, 2)$



(a) $A^\beta - (2, 1, 1)$



(a) $A^\beta - (2, 2, 2)$



(a) $A^\beta - (2, 5, 2)$



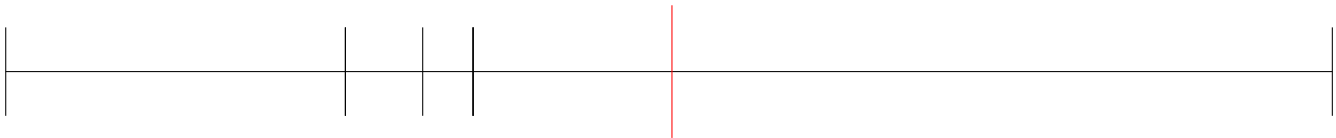
(a) $A^\beta - (3, 5, 2)$



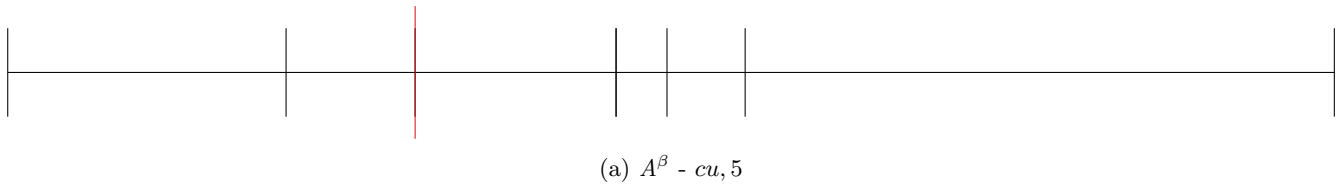
(a) $A^\beta - (4, 3, 2)$



(a) $A^\beta - (5, 4, 1)$



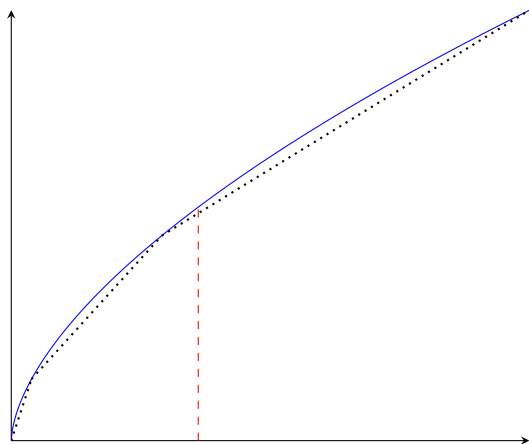
(a) $A^\beta - cu, 3$



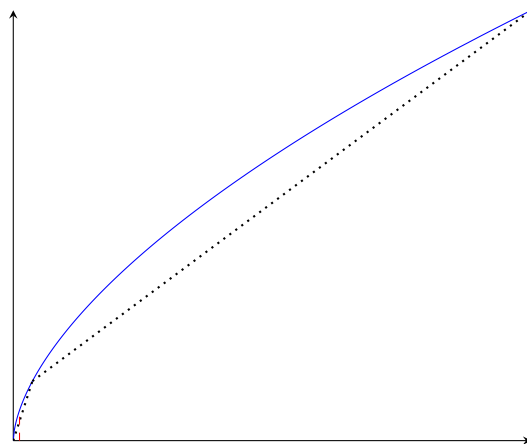
6 A^β Breakpoints

- (1, 1, 2)
 - q : 24.928068
 - $\hat{q}^\beta$: 6.705170
 - q^β : 6.886732
 - Breakpoints: [0, 2.7005882352941173, 20.05363555217427, 69.05411764705883]
 - Error: 0.181562217045779
 - Relative error: 0.026364060512248
- (2, 1, 1)
 - q : 1.077694
 - $\hat{q}^\beta$: 0.668187
 - q^β : 1.045917
 - Breakpoints: [0, 3.3036464983028657, 89.41176470588236]
 - Error: 0.377730099510029
 - Relative error: 0.361147172801027
- (2, 2, 2)
 - q : 41.494762
 - $\hat{q}^\beta$: 8.842280
 - q^β : 9.349663
 - Breakpoints: [0, 18.89081402008292, 76.14235294117645]
 - Error: 0.507383269265750
 - Relative error: 0.054267545119266
- (2, 5, 2)
 - q : 2.702329
 - $\hat{q}^\beta$: 1.550429
 - q^β : 1.815695
 - Breakpoints: [0, 4.0106608195978595, 138.54823529411763]
 - Error: 0.265266317738963
 - Relative error: 0.146096294252261
- (3, 5, 2)
 - q : 27.976182
 - $\hat{q}^\beta$: 6.746673
 - q^β : 7.380284
 - Breakpoints: [0, 8.941176470587976, 55.82566076118323, 181.68117647058827]
 - Error: 0.633610940996525
 - Relative error: 0.085851838058673
- (4, 3, 2)

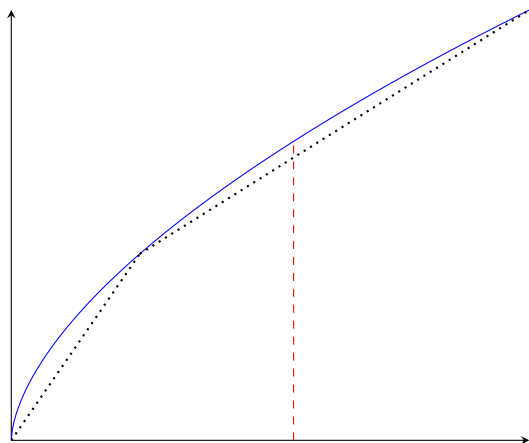
- q : 23.668655
- $\hat{q}^\beta$: 6.675812
- q^β : 6.675813
- Breakpoints: [0, 19.92063386984238, 23.668855770059224, 181.14352941176475]
- Error: 0.000001165600096
- Relative error: 0.000000174600479
- (5, 4, 1)
 - q : 82.708997
 - $\hat{q}^\beta$: 13.119333
 - q^β : 14.142678
 - Breakpoints: [0, 7.000222767756085, 12.878068451030797, 38.682379459353704, 192.3352941176471]
 - Error: 1.023345277940566
 - Relative error: 0.072358663567660
- $cu, 3$
 - q : 18.852086
 - $\hat{q}^\beta$: 5.668788
 - q^β : 5.823923
 - Breakpoints: [0, 7.499856372317694, 10.320104112088831, 13.6435791010403, 14.376341772690381, 16.73892021814917, 16.738997089244634, 16.738997089244645, 128.44338870924773]
 - Error: 0.155135739685360
 - Relative error: 0.026637668962737
- $cu, 5$
 - q : 16.235136
 - $\hat{q}^\beta$: 5.308038
 - q^β : 5.324435
 - Breakpoints: [0, 6.717165864481, 9.832103793393394, 14.682070969056184, 20.94487303718885, 20.944873037188852, 21.50633939847323, 22.410514202757604, 127.1036185995295]
 - Error: 0.016397064527812
 - Relative error: 0.003079587514971



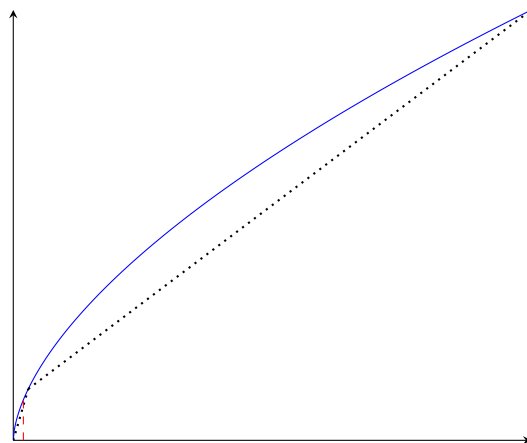
(a) $q^\beta - (1, 1, 2)$



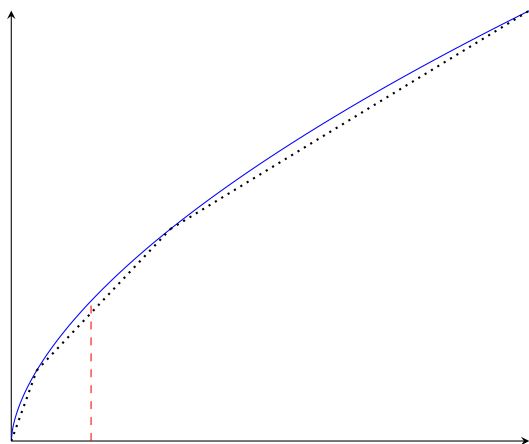
(b) $q^\beta - (2, 1, 1)$



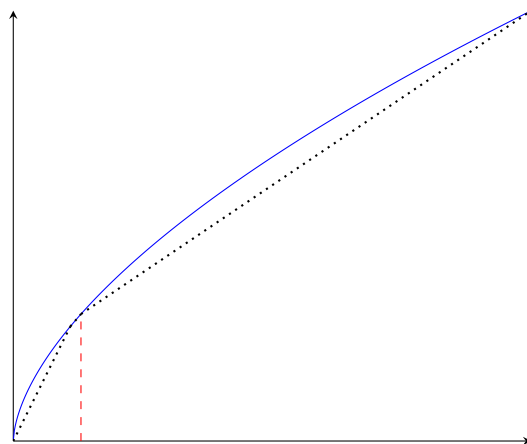
(c) $q^\beta - (2, 2, 2)$



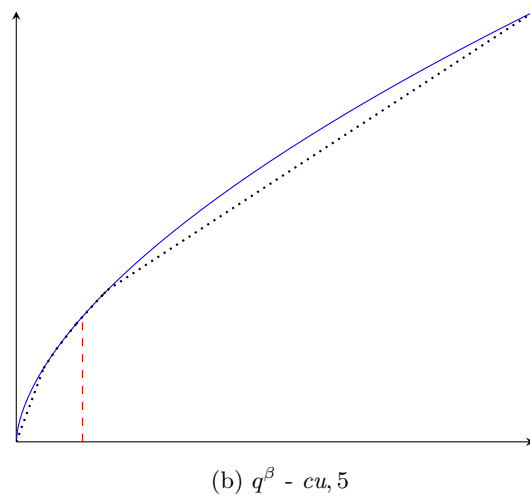
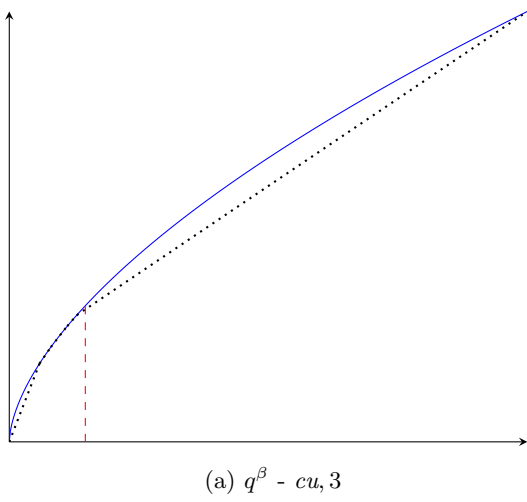
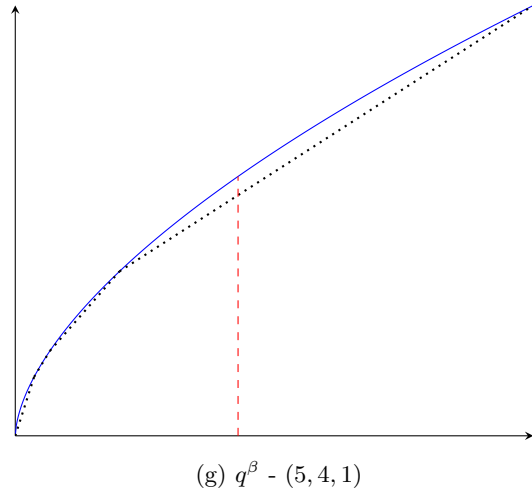
(d) $q^\beta - (2, 5, 2)$



(e) $q^\beta - (3, 5, 2)$



(f) $q^\beta - (4, 3, 2)$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	0.0	$z_{4,5,2}$	0.0	$A_{2,5,1}^\beta$	0.0
$z_{1,2,1}$	0.0	$z_{5,1,1}$	0.0	$A_{3,1,1}^\beta$	0.0
$z_{1,2,2}$	0.0	$z_{5,1,2}$	0.0	$A_{3,1,2}^\beta$	0.0
$z_{1,3,1}$	0.0	$z_{5,2,1}$	0.0	$A_{3,2,1}^\beta$	0.0
$z_{1,3,2}$	0.0	$z_{5,2,2}$	0.0	$A_{3,2,2}^\beta$	0.0
$z_{1,4,1}$	0.0	$z_{5,3,1}$	0.0	$A_{3,3,1}^\beta$	0.0
$z_{1,4,2}$	0.0	$z_{5,3,2}$	0.0	$A_{3,3,2}^\beta$	0.0
$z_{1,5,1}$	0.0	$z_{5,4,2}$	0.0	$A_{3,4,1}^\beta$	0.0
$z_{1,5,2}$	0.0	$z_{5,5,1}$	0.0	$A_{3,4,2}^\beta$	0.0
$z_{2,1,2}$	0.0	$z_{5,5,2}$	0.0	$A_{3,5,1}^\beta$	0.0
$z_{2,2,1}$	0.0	$z_{cu,1}$	0.0	$A_{4,1,1}^\beta$	0.0
$z_{2,3,1}$	0.0	$z_{cu,2}$	0.0	$A_{4,1,2}^\beta$	0.0
$z_{2,3,2}$	0.0	$z_{cu,4}$	0.0	$A_{4,2,1}^\beta$	0.0
$z_{2,4,1}$	0.0	$z_{hu,1}$	0.0	$A_{4,2,2}^\beta$	0.0
$z_{2,4,2}$	0.0	$z_{hu,2}$	0.0	$A_{4,3,1}^\beta$	0.0
$z_{2,5,1}$	0.0	$z_{hu,3}$	0.0	$A_{4,4,1}^\beta$	0.0
$z_{3,1,1}$	0.0	$z_{hu,4}$	0.0	$A_{4,4,2}^\beta$	0.0
$z_{3,1,2}$	0.0	$z_{hu,5}$	0.0	$A_{4,5,1}^\beta$	0.0
$z_{3,2,1}$	0.0	$A_{1,1,1}^\beta$	0.0	$A_{4,5,2}^\beta$	0.0
$z_{3,2,2}$	0.0	$A_{1,2,1}^\beta$	0.0	$A_{5,1,1}^\beta$	0.0
$z_{3,3,1}$	0.0	$A_{1,2,2}^\beta$	0.0	$A_{5,1,2}^\beta$	0.0
$z_{3,3,2}$	0.0	$A_{1,3,1}^\beta$	0.0	$A_{5,2,1}^\beta$	0.0
$z_{3,4,1}$	0.0	$A_{1,3,2}^\beta$	0.0	$A_{5,2,2}^\beta$	0.0
$z_{3,4,2}$	0.0	$A_{1,4,1}^\beta$	0.0	$A_{5,3,1}^\beta$	0.0
$z_{3,5,1}$	0.0	$A_{1,4,2}^\beta$	0.0	$A_{5,3,2}^\beta$	0.0
$z_{4,1,1}$	0.0	$A_{1,5,1}^\beta$	0.0	$A_{5,4,2}^\beta$	0.0
$z_{4,1,2}$	0.0	$A_{1,5,2}^\beta$	0.0	$A_{5,5,1}^\beta$	0.0
$z_{4,2,1}$	0.0	$A_{2,1,2}^\beta$	0.0	$A_{5,5,2}^\beta$	0.0
$z_{4,2,2}$	0.0	$A_{2,2,1}^\beta$	0.0	$A_{cu,1}^\beta$	0.0
$z_{4,3,1}$	0.0	$A_{2,3,1}^\beta$	0.0	$A_{cu,2}^\beta$	0.0
$z_{4,4,1}$	0.0	$A_{2,3,2}^\beta$	0.0	$A_{cu,4}^\beta$	0.0
$z_{4,4,2}$	0.0	$A_{2,4,1}^\beta$	0.0	$A_{hu,1}^\beta$	0.0
$z_{4,5,1}$	0.0	$A_{2,4,2}^\beta$	0.0	$A_{hu,2}^\beta$	0.0

Variable	Value	Variable	Value	Variable	Value
$A_{hu,3}^\beta$	0.0	$A_{4,4,1}$	0.0	$dt_{1,5,3}$	66.7
$A_{hu,4}^\beta$	0.0	$A_{4,4,2}$	0.0	$dt_{2,1,3}$	100.0
$A_{hu,5}^\beta$	0.0	$A_{4,5,1}$	0.0	$dt_{2,2,1}$	111.1
$A_{1,1,1}$	0.0	$A_{4,5,2}$	0.0	$dt_{2,3,1}$	111.1
$A_{1,2,1}$	0.0	$A_{5,1,1}$	0.0	$dt_{2,3,2}$	10.0
$A_{1,2,2}$	0.0	$A_{5,1,2}$	0.0	$dt_{2,3,3}$	183.3
$A_{1,3,1}$	0.0	$A_{5,2,1}$	0.0	$dt_{2,4,1}$	111.1
$A_{1,3,2}$	0.0	$A_{5,2,2}$	0.0	$dt_{2,4,2}$	166.7
$A_{1,4,1}$	0.0	$A_{5,3,1}$	0.0	$dt_{2,4,3}$	94.5
$A_{1,4,2}$	0.0	$A_{5,3,2}$	0.0	$dt_{2,5,1}$	111.1
$A_{1,5,1}$	0.0	$A_{5,4,2}$	0.0	$dt_{3,1,1}$	161.1
$A_{1,5,2}$	0.0	$A_{5,5,1}$	0.0	$dt_{3,1,2}$	166.7
$A_{2,1,2}$	0.0	$A_{5,5,2}$	0.0	$dt_{3,1,3}$	10.0
$A_{2,2,1}$	0.0	$A_{cu,1}$	0.0	$dt_{3,2,1}$	111.1
$A_{2,3,1}$	0.0	$A_{cu,2}$	0.0	$dt_{3,2,2}$	10.0
$A_{2,3,2}$	0.0	$A_{cu,4}$	0.0	$dt_{3,2,3}$	10.0
$A_{2,4,1}$	0.0	$A_{hu,1}$	0.0	$dt_{3,3,1}$	161.1
$A_{2,4,2}$	0.0	$A_{hu,2}$	0.0	$dt_{3,3,2}$	161.1
$A_{2,5,1}$	0.0	$A_{hu,3}$	0.0	$dt_{3,3,3}$	188.9
$A_{3,1,1}$	0.0	$A_{hu,4}$	0.0	$dt_{3,4,1}$	144.5
$A_{3,1,2}$	0.0	$A_{hu,5}$	0.0	$dt_{3,4,2}$	144.5
$A_{3,2,1}$	0.0	$dt_{1,1,1}$	66.7	$dt_{3,4,3}$	10.0
$A_{3,2,2}$	0.0	$dt_{1,2,1}$	17.15006133296988	$dt_{3,5,1}$	133.4
$A_{3,3,1}$	0.0	$dt_{1,2,2}$	17.15006133296988	$dt_{4,1,1}$	122.2
$A_{3,3,2}$	0.0	$dt_{1,2,3}$	17.15006133296988	$dt_{4,1,2}$	144.96842105263025
$A_{3,4,1}$	0.0	$dt_{1,3,1}$	66.7	$dt_{4,1,3}$	100.0
$A_{3,4,2}$	0.0	$dt_{1,3,2}$	66.7	$dt_{4,2,1}$	122.2
$A_{3,5,1}$	0.0	$dt_{1,3,3}$	122.2	$dt_{4,2,2}$	10.0
$A_{4,1,1}$	0.0	$dt_{1,4,1}$	66.7	$dt_{4,2,3}$	106.1
$A_{4,1,2}$	0.0	$dt_{1,4,2}$	10.0	$dt_{4,3,1}$	122.2
$A_{4,2,1}$	0.0	$dt_{1,4,3}$	77.8	$dt_{4,4,1}$	122.2
$A_{4,2,2}$	0.0	$dt_{1,5,1}$	66.7	$dt_{4,4,2}$	188.9
$A_{4,3,1}$	0.0	$dt_{1,5,2}$	66.7	$dt_{4,4,3}$	94.5

Variable	Value	Variable	Value	Variable	Value
$dt_{4,5,1}$	122.2	$f_{2,1,2}^H$	0.0	$f_{5,5,2}^H$	11.853112378346907
$dt_{4,5,2}$	122.2	$f_{2,2,1}^H$	0.0	$f_{1,1,1}^C$	0.0
$dt_{4,5,3}$	111.1	$f_{2,3,1}^H$	9.042484248424875	$f_{1,2,1}^C$	6.1
$dt_{5,1,1}$	133.3	$f_{2,3,2}^H$	0.0	$f_{1,2,2}^C$	0.0
$dt_{5,1,2}$	10.0	$f_{2,4,1}^H$	0.0	$f_{1,3,1}^C$	0.0
$dt_{5,1,3}$	10.0	$f_{2,4,2}^H$	0.0	$f_{1,3,2}^C$	0.0
$dt_{5,2,1}$	74.66004781348396	$f_{2,5,1}^H$	0.0	$f_{1,4,1}^C$	0.0
$dt_{5,2,2}$	10.0	$f_{3,1,1}^H$	0.0	$f_{1,4,2}^C$	0.0
$dt_{5,2,3}$	10.0	$f_{3,1,2}^H$	0.0	$f_{1,5,1}^C$	13.9
$dt_{5,3,1}$	133.3	$f_{3,2,1}^H$	14.8	$f_{1,5,2}^C$	0.0
$dt_{5,3,2}$	10.0	$f_{3,2,2}^H$	7.4326505276226005	$f_{2,1,2}^C$	0.0
$dt_{5,3,3}$	161.1	$f_{3,3,1}^H$	0.0	$f_{2,2,1}^C$	0.0
$dt_{5,4,3}$	24.335593220338993	$f_{3,3,2}^H$	0.0	$f_{2,3,1}^C$	0.0
$dt_{5,5,1}$	105.6	$f_{3,4,1}^H$	0.0	$f_{2,3,2}^C$	0.0
$dt_{5,5,2}$	10.0	$f_{3,4,2}^H$	0.0	$f_{2,4,1}^C$	0.0
$dt_{5,5,3}$	105.6	$f_{3,5,1}^H$	0.0	$f_{2,4,2}^C$	0.0
$dt_{cu,1}$	53.3	$f_{4,1,1}^H$	12.6	$f_{2,5,1}^C$	0.0
$dt_{cu,2}$	97.8	$f_{4,1,2}^H$	0.0	$f_{3,1,1}^C$	0.0
$dt_{cu,4}$	108.9	$f_{4,2,1}^H$	0.0	$f_{3,1,2}^C$	1.7303999999999996
$dt_{hu,1}$	80.0	$f_{4,2,2}^H$	0.0	$f_{3,2,1}^C$	0.0
$dt_{hu,2}$	18.3	$f_{4,3,1}^H$	0.0	$f_{3,2,2}^C$	0.0
$dt_{hu,3}$	18.9	$f_{4,4,1}^H$	0.0	$f_{3,3,1}^C$	0.0
$dt_{hu,4}$	63.3	$f_{4,4,2}^H$	0.0	$f_{3,3,2}^C$	0.0
$dt_{hu,5}$	35.6	$f_{4,5,1}^H$	0.0	$f_{3,4,1}^C$	0.0
$f_{1,1,1}^H$	8.8	$f_{4,5,2}^H$	0.0	$f_{3,4,2}^C$	17.3
$f_{1,2,1}^H$	0.0	$f_{5,1,1}^H$	0.0	$f_{3,5,1}^C$	0.0
$f_{1,2,2}^H$	0.0	$f_{5,1,2}^H$	5.846887621653093	$f_{4,1,1}^C$	0.0
$f_{1,3,1}^H$	0.0	$f_{5,2,1}^H$	5.435558889722436	$f_{4,1,2}^C$	0.0
$f_{1,3,2}^H$	0.0	$f_{5,2,2}^H$	0.0	$f_{4,2,1}^C$	0.0
$f_{1,4,1}^H$	0.0	$f_{5,3,1}^H$	0.0	$f_{4,2,2}^C$	0.0
$f_{1,4,2}^H$	0.0	$f_{5,3,2}^H$	0.0	$f_{4,3,1}^C$	8.4
$f_{1,5,1}^H$	0.0	$f_{5,4,2}^H$	0.0	$f_{4,4,1}^C$	0.0
$f_{1,5,2}^H$	0.0	$f_{5,5,1}^H$	0.0	$f_{4,4,2}^C$	0.0

Variable	Value	Variable	Value	Variable	Value
$f_{4,5,1}^C$	0.0	$\text{RecLMTD}_{3,4,1}$	0.0	$d_{4,3,2}$	0.0
$f_{4,5,2}^C$	0.0	$\text{RecLMTD}_{3,4,2}$	0.0	$d_{4,4,1}$	0.0
$f_{5,1,1}^C$	0.0	$\text{RecLMTD}_{3,5,1}$	0.013119967177547672	$d_{4,4,2}$	0.0
$f_{5,1,2}^C$	0.0	$\text{RecLMTD}_{4,1,1}$	0.020350921041481694	$d_{4,5,1}$	0.0
$f_{5,2,1}^C$	0.0	$\text{RecLMTD}_{4,1,2}$	0.02150212335787724	$d_{4,5,2}$	0.0
$f_{5,2,2}^C$	0.0	$\text{RecLMTD}_{4,2,1}$	0.0	$d_{4,1,2}$	0.0
$f_{5,3,1}^C$	0.0	$\text{RecLMTD}_{4,2,2}$	0.029297885186065308	$d_{4,2,1}$	0.0
$f_{5,3,2}^C$	0.0	$\text{RecLMTD}_{4,3,1}$	0.012328501480204937	$d_{4,3,1}$	0.0
$f_{5,4,2}^C$	0.0	$\text{RecLMTD}_{4,4,1}$	0.02230651890829312	$d_{4,3,2}$	0.0
$f_{5,5,1}^C$	0.0	$\text{RecLMTD}_{4,4,2}$	0.0066165571987497375	$d_{4,4,1}$	0.0
$f_{5,5,2}^C$	0.0	$\text{RecLMTD}_{4,5,1}$	0.007439995191222409	$d_{4,4,2}$	0.0
$\text{RecLMTD}_{1,1,1}$	0.1	$\text{RecLMTD}_{4,5,2}$	0.02554670259920889	$d_{4,5,1}$	0.0
$\text{RecLMTD}_{1,2,1}$	0.1	$\text{RecLMTD}_{5,1,1}$	0.0	$d_{4,3,1,1}$	0.0
$\text{RecLMTD}_{1,2,2}$	0.1	$\text{RecLMTD}_{5,1,2}$	0.0	$d_{4,3,1,2}$	0.0
$\text{RecLMTD}_{1,3,1}$	0.1	$\text{RecLMTD}_{5,2,1}$	0.0	$d_{4,3,2,1}$	0.0
$\text{RecLMTD}_{1,3,2}$	0.1	$\text{RecLMTD}_{5,2,2}$	0.0	$d_{4,3,2,2}$	0.0
$\text{RecLMTD}_{1,4,1}$	0.1	$\text{RecLMTD}_{5,3,1}$	0.0	$d_{4,3,3,1}$	0.0
$\text{RecLMTD}_{1,4,2}$	0.1	$\text{RecLMTD}_{5,3,2}$	0.028888218232147534	$d_{4,3,3,2}$	0.0
$\text{RecLMTD}_{1,5,1}$	0.1	$\text{RecLMTD}_{5,4,2}$	0.04141640783887002	$d_{4,3,4,1}$	0.0
$\text{RecLMTD}_{1,5,2}$	0.1	$\text{RecLMTD}_{5,5,1}$	0.0	$d_{4,3,4,2}$	0.0
$\text{RecLMTD}_{2,1,2}$	0.1	$\text{RecLMTD}_{5,5,2}$	0.0	$d_{4,3,5,1}$	0.0
$\text{RecLMTD}_{2,2,1}$	0.1	$\text{RecLMTD}_{cu,1}$	0.016108908592626668	$d_{4,1,1,1}$	0.0
$\text{RecLMTD}_{2,3,1}$	0.1	$\text{RecLMTD}_{cu,2}$	0.009423141115769086	$d_{4,1,1,2}$	0.0
$\text{RecLMTD}_{2,3,2}$	0.020926522199679467	$\text{RecLMTD}_{cu,4}$	0.008535688158765427	$d_{4,2,1}$	0.0
$\text{RecLMTD}_{2,4,1}$	0.1	$\text{RecLMTD}_{hu,1}$	0.012387051800747542	$d_{4,2,2}$	0.0
$\text{RecLMTD}_{2,4,2}$	0.007297880555862293	$\text{RecLMTD}_{hu,2}$	0.045618305265862594	$d_{4,3,1}$	0.0
$\text{RecLMTD}_{2,5,1}$	0.1	$\text{RecLMTD}_{hu,3}$	0.04305097282506776	$d_{4,4,1}$	0.0
$\text{RecLMTD}_{3,1,1}$	0.1	$\text{RecLMTD}_{hu,4}$	0.015434474530129737	$d_{4,4,2}$	0.0
$\text{RecLMTD}_{3,1,2}$	0.1	$\text{RecLMTD}_{hu,5}$	0.024141568940966463	$d_{4,5,1}$	0.0
$\text{RecLMTD}_{3,2,1}$	0.1	$q_{1,1,1}$	0.0	$d_{4,5,2}$	0.0
$\text{RecLMTD}_{3,2,2}$	0.1	$q_{1,2,1}$	0.0	$d_{4,1,1,1}$	0.0
$\text{RecLMTD}_{3,3,1}$	0.1	$q_{1,2,2}$	0.0	$d_{4,1,1,2}$	0.0
$\text{RecLMTD}_{3,3,2}$	0.026450800778310225	$q_{1,3,1}$	0.0	$d_{4,2,1}$	0.0

Variable	Value	Variable	Value	Variable	Value
$q_{5,2,2}$	0.0	$t_{3,2,2}^H$	163.34633007007847	$t_{2,1,2}^C$	129.79605263157896
$q_{5,3,1}$	0.0	$t_{3,3,1}^H$	226.7	$t_{2,2,1}^C$	221.7
$q_{5,3,2}$	0.0	$t_{3,3,2}^H$	226.7	$t_{2,3,1}^C$	221.1
$q_{5,4,2}$	0.0	$t_{3,4,1}^H$	226.7	$t_{2,3,2}^C$	37.8
$q_{5,5,1}$	0.0	$t_{3,4,2}^H$	226.7	$t_{2,4,1}^C$	86.51229399007087
$q_{5,5,2}$	0.0	$t_{3,5,1}^H$	85.59536151261509	$t_{2,4,2}^C$	112.86184971098264
$q_{cu,1}$	0.0	$t_{4,1,1}^H$	271.1	$t_{2,5,1}^C$	204.4
$q_{cu,2}$	0.0	$t_{4,1,2}^H$	271.1	$t_{3,1,1}^C$	160.0
$q_{cu,4}$	0.0	$t_{4,2,1}^H$	271.1	$t_{3,1,2}^C$	60.0
$q_{hu,1}$	0.0	$t_{4,2,2}^H$	271.1	$t_{3,2,1}^C$	221.7
$q_{hu,2}$	0.0	$t_{4,3,1}^H$	271.1	$t_{3,2,2}^C$	115.6
$q_{hu,3}$	0.0	$t_{4,4,1}^H$	271.1	$t_{3,3,1}^C$	221.1
$q_{hu,4}$	0.0	$t_{4,4,2}^H$	271.1	$t_{3,3,2}^C$	37.8
$q_{hu,5}$	0.0	$t_{4,5,1}^H$	271.1	$t_{3,4,1}^C$	104.72432380851744
$t_{1,1,1}^H$	160.0	$t_{4,5,2}^H$	271.1	$t_{3,4,2}^C$	82.2
$t_{1,2,1}^H$	160.0	$t_{5,1,1}^H$	198.9	$t_{3,5,1}^C$	204.4
$t_{1,2,2}^H$	160.0	$t_{5,1,2}^H$	65.6	$t_{4,1,1}^C$	160.0
$t_{1,3,1}^H$	160.0	$t_{5,2,1}^H$	198.9	$t_{4,1,2}^C$	60.0
$t_{1,3,2}^H$	160.0	$t_{5,2,2}^H$	102.51642558256174	$t_{4,2,1}^C$	221.7
$t_{1,4,1}^H$	160.0	$t_{5,3,1}^H$	198.9	$t_{4,2,2}^C$	115.6
$t_{1,4,2}^H$	119.38522727272726	$t_{5,3,2}^H$	65.6	$t_{4,3,1}^C$	221.1
$t_{1,5,1}^H$	160.0	$t_{5,4,2}^H$	65.6	$t_{4,4,1}^C$	104.72432380851744
$t_{1,5,2}^H$	160.0	$t_{5,5,1}^H$	198.9	$t_{4,4,2}^C$	82.2
$t_{2,1,2}^H$	137.8	$t_{5,5,2}^H$	126.72824858757063	$t_{4,5,1}^C$	93.3
$t_{2,2,1}^H$	160.34047781513647	$t_{1,1,1}^C$	90.20394736842101	$t_{4,5,2}^C$	93.3
$t_{2,3,1}^H$	248.9	$t_{1,2,1}^C$	221.7	$t_{5,1,1}^C$	92.62135922330096
$t_{2,3,2}^H$	137.8	$t_{1,2,2}^C$	115.6	$t_{5,1,2}^C$	60.0
$t_{2,4,1}^H$	148.8150943396225	$t_{1,3,1}^C$	221.1	$t_{5,2,1}^C$	221.7
$t_{2,4,2}^H$	137.8	$t_{1,3,2}^C$	215.35238095238094	$t_{5,2,2}^C$	115.6
$t_{2,5,1}^H$	198.85754716981077	$t_{1,4,1}^C$	102.85953757225435	$t_{5,3,1}^C$	221.1
$t_{3,1,1}^H$	123.99729729729725	$t_{1,4,2}^C$	82.2	$t_{5,3,2}^C$	37.8
$t_{3,1,2}^H$	116.23716216216215	$t_{1,5,1}^C$	204.4	$t_{5,4,2}^C$	146.03815028901732
$t_{3,2,1}^H$	226.7	$t_{1,5,2}^C$	93.3	$t_{5,5,1}^C$	93.3

Variable	Value	Variable	Value	Variable	Value
$t_{5,5,2}^C$	93.3	$b_{4,5,1}^{H,in}$	0.0	$b_{3,4,1}^{H,out}$	0.0
$b_{1,1,1}^{H,in}$	1408.0	$b_{4,5,2}^{H,in}$	0.0	$b_{3,4,2}^{H,out}$	0.0
$b_{1,2,1}^{H,in}$	0.0	$b_{5,1,1}^{H,in}$	0.0	$b_{3,5,1}^{H,out}$	0.0
$b_{1,2,2}^{H,in}$	0.0	$b_{5,1,2}^{H,in}$	383.555827980443	$b_{4,1,1}^{H,out}$	3415.86
$b_{1,3,1}^{H,in}$	0.0	$b_{5,2,1}^{H,in}$	1081.1326631657926	$b_{4,1,2}^{H,out}$	0.0
$b_{1,3,2}^{H,in}$	0.0	$b_{5,2,2}^{H,in}$	0.0	$b_{4,2,1}^{H,out}$	0.0
$b_{1,4,1}^{H,in}$	0.0	$b_{5,3,1}^{H,in}$	0.0	$b_{4,2,2}^{H,out}$	0.0
$b_{1,4,2}^{H,in}$	0.0	$b_{5,3,2}^{H,in}$	0.0	$b_{4,3,1}^{H,out}$	0.0
$b_{1,5,1}^{H,in}$	0.0	$b_{5,4,2}^{H,in}$	0.0	$b_{4,4,1}^{H,out}$	0.0
$b_{1,5,2}^{H,in}$	0.0	$b_{5,5,1}^{H,in}$	0.0	$b_{4,4,2}^{H,out}$	0.0
$b_{2,1,2}^{H,in}$	0.0	$b_{5,5,2}^{H,in}$	1502.1241720195574	$b_{4,5,1}^{H,out}$	0.0
$b_{2,2,1}^{H,in}$	0.0	$b_{1,1,1}^{H,out}$	1408.0	$b_{4,5,2}^{H,out}$	0.0
$b_{2,3,1}^{H,in}$	2250.6743294329517	$b_{1,2,1}^{H,out}$	0.0	$b_{5,1,1}^{H,out}$	0.0
$b_{2,3,2}^{H,in}$	0.0	$b_{1,2,2}^{H,out}$	0.0	$b_{5,1,2}^{H,out}$	383.555827980443
$b_{2,4,1}^{H,in}$	0.0	$b_{1,3,1}^{H,out}$	0.0	$b_{5,2,1}^{H,out}$	1081.1326631657926
$b_{2,4,2}^{H,in}$	0.0	$b_{1,3,2}^{H,out}$	0.0	$b_{5,2,2}^{H,out}$	0.0
$b_{2,5,1}^{H,in}$	0.0	$b_{1,4,1}^{H,out}$	0.0	$b_{5,3,1}^{H,out}$	0.0
$b_{3,1,1}^{H,in}$	0.0	$b_{1,4,2}^{H,out}$	0.0	$b_{5,3,2}^{H,out}$	0.0
$b_{3,1,2}^{H,in}$	0.0	$b_{1,5,1}^{H,out}$	0.0	$b_{5,4,2}^{H,out}$	0.0
$b_{3,2,1}^{H,in}$	3355.16	$b_{1,5,2}^{H,out}$	0.0	$b_{5,5,1}^{H,out}$	0.0
$b_{3,2,2}^{H,in}$	1684.9818746120434	$b_{2,1,2}^{H,out}$	0.0	$b_{5,5,2}^{H,out}$	1502.1241720195574
$b_{3,3,1}^{H,in}$	0.0	$b_{2,2,1}^{H,out}$	0.0	$b_{1,1,1}^{C,in}$	0.0
$b_{3,3,2}^{H,in}$	0.0	$b_{2,3,1}^{H,out}$	2250.6743294329517	$b_{1,2,1}^{C,in}$	1352.37
$b_{3,4,1}^{H,in}$	0.0	$b_{2,3,2}^{H,out}$	0.0	$b_{1,2,2}^{C,in}$	0.0
$b_{3,4,2}^{H,in}$	0.0	$b_{2,4,1}^{H,out}$	0.0	$b_{1,3,1}^{C,in}$	0.0
$b_{3,5,1}^{H,in}$	0.0	$b_{2,4,2}^{H,out}$	0.0	$b_{1,3,2}^{C,in}$	0.0
$b_{4,1,1}^{H,in}$	3415.86	$b_{2,5,1}^{H,out}$	0.0	$b_{1,4,1}^{C,in}$	0.0
$b_{4,1,2}^{H,in}$	0.0	$b_{3,1,1}^{H,out}$	0.0	$b_{1,4,2}^{C,in}$	0.0
$b_{4,2,1}^{H,in}$	0.0	$b_{3,1,2}^{H,out}$	0.0	$b_{1,5,1}^{C,in}$	2841.1600000000003
$b_{4,2,2}^{H,in}$	0.0	$b_{3,2,1}^{H,out}$	3355.16	$b_{1,5,2}^{C,in}$	0.0
$b_{4,3,1}^{H,in}$	0.0	$b_{3,2,2}^{H,out}$	1684.9818746120434	$b_{2,1,2}^{C,in}$	0.0
$b_{4,4,1}^{H,in}$	0.0	$b_{3,3,1}^{H,out}$	0.0	$b_{2,2,1}^{C,in}$	0.0
$b_{4,4,2}^{H,in}$	0.0	$b_{3,3,2}^{H,out}$	0.0	$b_{2,3,1}^{C,in}$	0.0

Variable	Value	Variable	Value	Variable	Value
$b_{2,3,2}^{C,in}$	0.0	$b_{1,2,2}^{C,out}$	0.0	$b_{5,1,2}^{C,out}$	0.0
$b_{2,4,1}^{C,in}$	0.0	$b_{1,3,1}^{C,out}$	0.0	$b_{5,2,1}^{C,out}$	0.0
$b_{2,4,2}^{C,in}$	0.0	$b_{1,3,2}^{C,out}$	0.0	$b_{5,2,2}^{C,out}$	0.0
$b_{2,5,1}^{C,in}$	0.0	$b_{1,4,1}^{C,out}$	0.0	$b_{5,3,1}^{C,out}$	0.0
$b_{3,1,1}^{C,in}$	0.0	$b_{1,4,2}^{C,out}$	0.0	$b_{5,3,2}^{C,out}$	0.0
$b_{3,1,2}^{C,in}$	103.82399999999976	$b_{1,5,1}^{C,out}$	2841.16000000000003	$b_{5,4,2}^{C,out}$	0.0
$b_{3,2,1}^{C,in}$	0.0	$b_{1,5,2}^{C,out}$	0.0	$b_{5,5,1}^{C,out}$	0.0
$b_{3,2,2}^{C,in}$	0.0	$b_{2,1,2}^{C,out}$	0.0	$b_{5,5,2}^{C,out}$	0.0
$b_{3,3,1}^{C,in}$	0.0	$b_{2,2,1}^{C,out}$	0.0		
$b_{3,3,2}^{C,in}$	0.0	$b_{2,3,1}^{C,out}$	0.0		
$b_{3,4,1}^{C,in}$	0.0	$b_{2,3,2}^{C,out}$	0.0		
$b_{3,4,2}^{C,in}$	1422.06000000000002	$b_{2,4,1}^{C,out}$	0.0		
$b_{3,5,1}^{C,in}$	0.0	$b_{2,4,2}^{C,out}$	0.0		
$b_{4,1,1}^{C,in}$	0.0	$b_{2,5,1}^{C,out}$	0.0		
$b_{4,1,2}^{C,in}$	0.0	$b_{3,1,1}^{C,out}$	0.0		
$b_{4,2,1}^{C,in}$	0.0	$b_{3,1,2}^{C,out}$	103.82399999999976		
$b_{4,2,2}^{C,in}$	0.0	$b_{3,2,1}^{C,out}$	0.0		
$b_{4,3,1}^{C,in}$	1857.24	$b_{3,2,2}^{C,out}$	0.0		
$b_{4,4,1}^{C,in}$	0.0	$b_{3,3,1}^{C,out}$	0.0		
$b_{4,4,2}^{C,in}$	0.0	$b_{3,3,2}^{C,out}$	0.0		
$b_{4,5,1}^{C,in}$	0.0	$b_{3,4,1}^{C,out}$	0.0		
$b_{4,5,2}^{C,in}$	0.0	$b_{3,4,2}^{C,out}$	1422.06000000000002		
$b_{5,1,1}^{C,in}$	0.0	$b_{3,5,1}^{C,out}$	0.0		
$b_{5,1,2}^{C,in}$	0.0	$b_{4,1,1}^{C,out}$	0.0		
$b_{5,2,1}^{C,in}$	0.0	$b_{4,1,2}^{C,out}$	0.0		
$b_{5,2,2}^{C,in}$	0.0	$b_{4,2,1}^{C,out}$	0.0		
$b_{5,3,1}^{C,in}$	0.0	$b_{4,2,2}^{C,out}$	0.0		
$b_{5,3,2}^{C,in}$	0.0	$b_{4,3,1}^{C,out}$	1857.24		
$b_{5,4,2}^{C,in}$	0.0	$b_{4,4,1}^{C,out}$	0.0		
$b_{5,5,1}^{C,in}$	0.0	$b_{4,4,2}^{C,out}$	0.0		
$b_{5,5,2}^{C,in}$	0.0	$b_{4,5,1}^{C,out}$	0.0		
$b_{1,1,1}^{C,out}$	0.0	$b_{4,5,2}^{C,out}$	0.0		
$b_{1,2,1}^{C,out}$	1352.37	$b_{5,1,1}^{C,out}$	0.0		